

TASK 19: #51611

Learn about different tuning methods

1. Pagespeed
 2. Gzip compression
 3. tuned daemon
- etc

1. PageSpeed Check

Explanation: Tests how fast your webpage loads and gives suggestions to improve it.

Step 1: Install Apache (httpd)

```
sudo yum install httpd -y
sudo systemctl enable --now httpd
```

Step 2: Create a test webpage

```
sudo vi /var/www/html/index.html

inside:

<!DOCTYPE html>
<html lang="en">
<head>

  <meta charset="UTF-8">

  <meta name="viewport" content="width=device-width, initial-scale=1.0">

  <title>Server Tuning Test Page</title>
<style>

  * { margin: 0; padding: 0; box-sizing: border-box; }

  body {

    font-family: -apple-system, BlinkMacSystemFont, 'Segoe UI', Roboto, Arial,
sans-serif;

    background: #0f172a;

    color: #e2e8f0;

    display: flex;

    align-items: center;

    justify-content: center;
```

```
    min-height: 100vh;

    padding: 20px;
}

.card {

    background: #1e293b;

    border-radius: 12px;

    padding: 40px;

    max-width: 600px;

    width: 100%;

    box-shadow: 0 10px 30px rgba(0,0,0,0.4);
}

h1 {

    font-size: 28px;

    margin-bottom: 10px;

    color: #38bdf8;
}

p {

    line-height: 1.6;

    margin-bottom: 15px;

    color: #cbd5e1;
}

.badge {

    display: inline-block;

    background: #334155;

    color: #94a3b8;

    padding: 4px 12px;

    border-radius: 20px;

    font-size: 13px;

    margin: 4px 4px 4px 0;
}
```

```
.footer {
    margin-top: 20px;
    font-size: 13px;
    color: #64748b;
}

button {
    background: #38bdf8;
    color: #0f172a;
    border: none;
    padding: 10px 20px;
    border-radius: 6px;
    font-weight: 600;
    cursor: pointer;
    margin-top: 10px;
}

button:hover { background: #0ea5e9; }

</style>
</head>
<body>
    <div class="card">
        <h1>✔ Server Test Page</h1>

        <p>This static page is used to verify server configuration: gzip compression,
        caching headers, TLS setup, and general response speed on an AWS EC2 instance.</p>

        <div>
            <span class="badge">Nginx / Apache</span>
            <span class="badge">Gzip Compression</span>
            <span class="badge">Tuned Daemon</span>
            <span class="badge">AWS EC2</span>
        </div>
    </div>
```

```

<p id="time" class="footer"></p>

<button onclick="checkTime()">Check Page Load Time</button>

</div>

<script>

function checkTime() {

    const perf = performance.getEntriesByType("navigation")[0];

    const loadTime = perf ? perf.duration.toFixed(2) : "N/A";

    document.getElementById("time").innerText = "Page load time: " + loadTime + "
ms";

}

window.onload = checkTime;

</script>

</body>

</html>

```

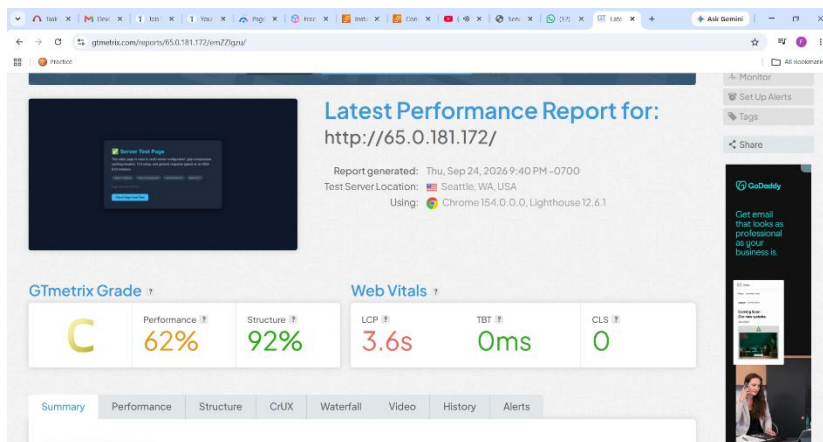
Step 3: Open the online tool

Go to <https://gtmetrix.web.dev> in your browser.

Step 4: Enter your site URL

<http://65.0.181.172/>

output:



2. Gzip Compression Setup & Check

Explanation: Compresses server responses (HTML/CSS/JS) so pages load faster and use less bandwidth.

Step 1: Create/edit deflate config (Apache)

```
sudo vi /etc/httpd/conf.d/deflate.conf
```

Step 2: Add compression rules

```
<IfModule mod_deflate.c>
  AddOutputFilterByType DEFLATE text/html text/plain text/css text/javascript
  application/javascript application/json application/xml
  DeflateCompressionLevel 6
</IfModule>
```

Step 3: Test and restart Apache

```
sudo httpd -t
sudo systemctl restart httpd
```

Step 4: Verify gzip is active

```
curl -H "Accept-Encoding: gzip" -I http://65.0.181.172/
```

Step 5: Confirm response header

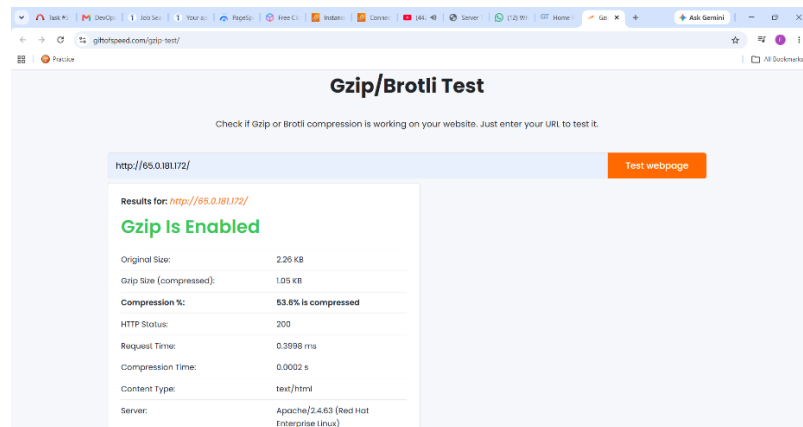
Look for: Content-Encoding: gzip

Step 6: Check with an online gzip tool

Visit <https://www.giftofspeed.com/gzip-test/> and enter your URL

<http://65.0.181.172/>

OUTPUT:



The screenshot shows a web browser displaying the 'Gzip/Brotli Test' page. The page has a light blue header with the title 'Gzip/Brotli Test' and a subtitle 'Check if Gzip or Brotli compression is working on your website. Just enter your URL to test it.' Below the header, there is a text input field containing 'http://65.0.181.172/' and an orange 'Test webpage' button. The results are displayed in a white box with a green header 'Results for: http://65.0.181.172/' and a green status 'Gzip Is Enabled'. The results table shows the following data:

Original Size:	2.26 KB
Gzip Size (compressed):	1.05 KB
Compression %:	53.6% is compressed
HTTP Status:	200
Request Time:	0.3998 ms
Compression Time:	0.0002 s
Content Type:	text/html
Server:	Apache/2.4.63 (Red Hat Enterprise Linux)

3. Tuned Daemon Setup & Check

Explanation: A background service that automatically tunes Linux kernel/CPU/network settings for better server performance.

Step 1: Install tuned

```
sudo yum install tuned -y
```

Step 2: Enable and start the service

```
sudo systemctl enable --now tuned
```

Step 3: Check service is running

```
sudo systemctl status tuned
```

Step 4: List available profiles

```
tuned-adm list
```

Step 5: Apply the AWS-optimized profile

```
sudo tuned-adm profile aws
```

Step 6: Confirm active profile

```
tuned-adm active
```

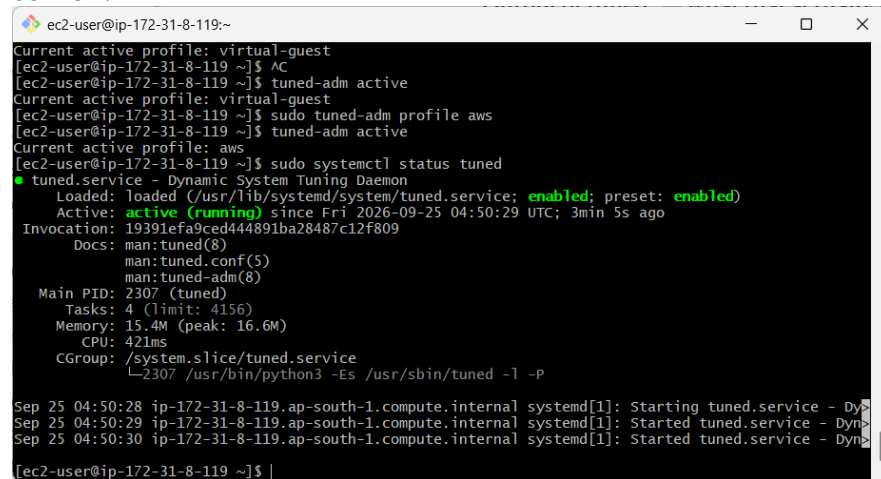
Step 7: View settings the profile applies

```
sudo tuned-adm profile_info aws
```

Step 8: Verify settings actually match the profile

```
sudo tuned-adm verify
```

OUTPUT:



```
ec2-user@ip-172-31-8-119:~  
Current active profile: virtual-guest  
[ec2-user@ip-172-31-8-119 ~]$ AC  
[ec2-user@ip-172-31-8-119 ~]$ tuned-adm active  
Current active profile: virtual-guest  
[ec2-user@ip-172-31-8-119 ~]$ sudo tuned-adm profile aws  
[ec2-user@ip-172-31-8-119 ~]$ tuned-adm active  
Current active profile: aws  
[ec2-user@ip-172-31-8-119 ~]$ sudo systemctl status tuned  
● tuned.service - Dynamic System Tuning Daemon  
   Loaded: loaded (/usr/lib/systemd/system/tuned.service; enabled; preset: enabled)  
   Active: active (running) since Fri 2026-09-25 04:50:29 UTC; 3min 5s ago  
 Invocation: 19391efa9ced444891ba28487c12f809  
    Docs: man:tuned(8)  
          man:tuned.conf(5)  
          man:tuned-adm(8)  
 Main PID: 2307 (tuned)  
   Tasks: 4 (limit: 4156)  
  Memory: 15.4M (peak: 16.6M)  
     CPU: 421ms  
    CGroup: /system.slice/tuned.service  
            └─2307 /usr/bin/python3 -Es /usr/sbin/tuned -l -P  
  
Sep 25 04:50:28 ip-172-31-8-119.ap-south-1.compute.internal systemd[1]: Starting tuned.service - Dyn  
Sep 25 04:50:29 ip-172-31-8-119.ap-south-1.compute.internal systemd[1]: Started tuned.service - Dyn  
Sep 25 04:50:30 ip-172-31-8-119.ap-south-1.compute.internal systemd[1]: Started tuned.service - Dyn  
[ec2-user@ip-172-31-8-119 ~]$
```